



Writing a Research Proposal

Research Seminar

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What is a research proposal?

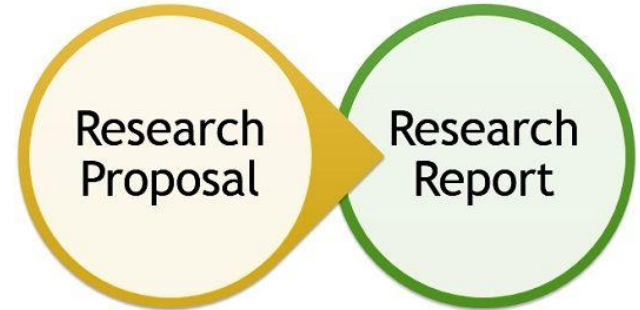
A research proposal is a formal document that explains:

- What you plan to research
- Why it is worth researching
- How you will do it



Research Proposal vs Research Report (Thesis)

- Research Proposal
 - Prepared before the research commences
 - Introduces the research problem, the motivation behind it and how to do the research
- Research Report (Thesis)
 - Prepared during and after the research work
 - Introduces the research problem, motivation, motivation, methods, results and discussion



About writing research proposals

- Before you begin with a research, you are required to submit a research proposal.
- By reviewing your proposal, the faculty members will accept or deny the research.
- You need to present a strong project proposal in order to get the approval and get started on the research.
- Topic needs to be specific with a well defined scope.
 - What is being investigated
 - Who/What it involves
 - In what context



Components of a research proposal

Following sections should be included in your project proposal:

1. **Abstract**
2. **Introduction**
 - a. Background
 - b. Motivation
 - c. Problem statement
 - d. Research Objectives
 - e. Preliminary Suppositions and Implications
3. Literature Review
4. Proposed method
5. Schedule/timeline
6. Conclusion
7. Bibliography



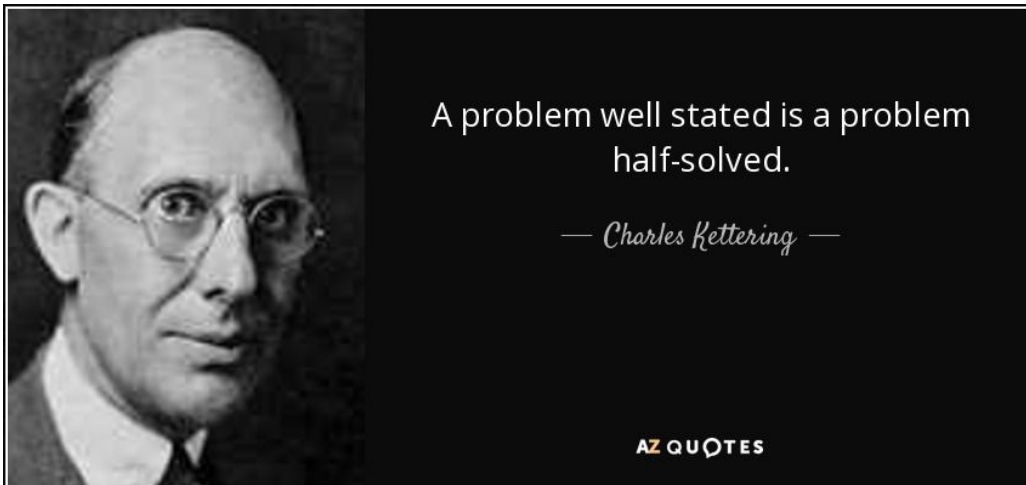
Abstract

Abstract

A standalone short paragraph (~300 words) that contains the following information

- Motivation
 - Why the reader might want to read it
- What is the problem you are trying to solve
- Approach
- Expected outcomes
- Concluding remark

1. Introduction



A problem well stated is a problem
half-solved.

— *Charles Kettering* —

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Introduction

The introduction is a narrative that briefly & clearly answers the following questions:

- What is the central research problem?
- What is the topic of study related to that problem?
- What makes it unique? (novelty)
- What methods should be used to analyze the research problem?
- Why this is an important research, what is its significance, and why should someone reading the proposal care about the outcomes of the proposed study? (motivation)
- What are your exact objectives of this research?
- In some cases, commercial value.

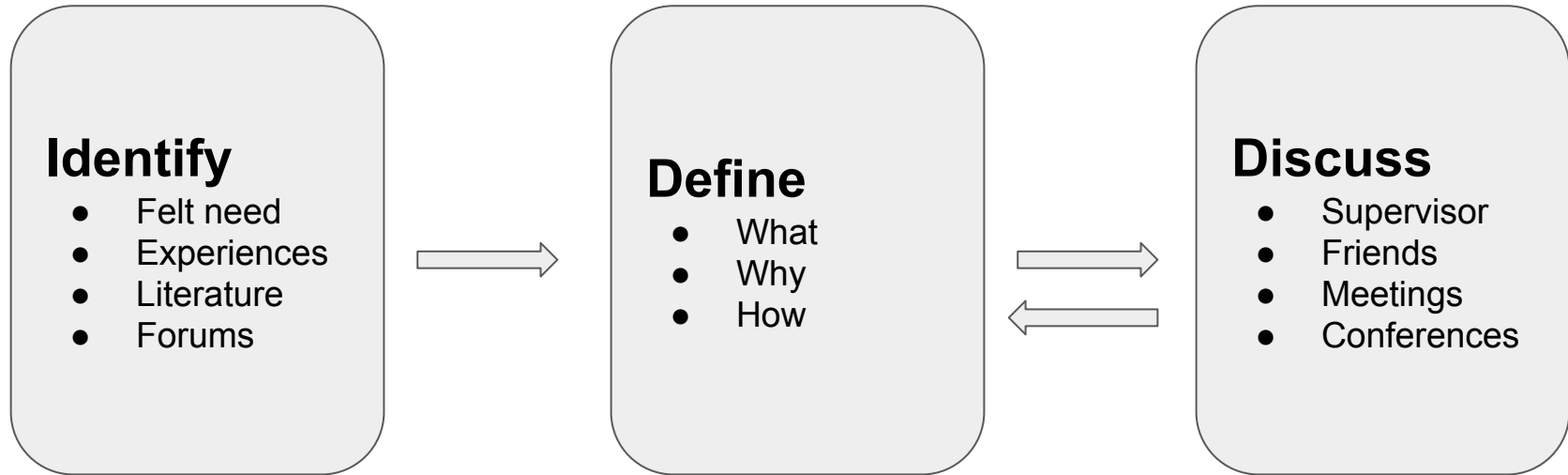
Background & Motivation

- This is where you explain the context of your proposal and describe in detail why it's important (motivation).
- State the **research problem** and give a more detailed explanation of the purpose of the study
- Present the rationale of your proposed study and clearly indicate why it is worth doing.
 - Answer the "So What? question [Why should anyone care?].
- Describe the major issues or problems to be addressed by your research.
- Note how your proposed study builds on previous work about the research problem.

Defining the Research Problem

- A clear statement of what challenge/problem you are trying to address
- Better if it addresses a felt need
 - Can do just for a fun of doing research
- If you clearly understand the problem, half the problem is solved!

Problem formulation



Characteristics of a good problem

- Interesting
 - Keep researcher interested in it throughout the research process
 - Will it lead to generation of new knowledge, understanding, or tools?
 - What others think about it?
- Relevant
 - Is the problem significant enough in today's context?
- Clear (unambiguous)
 - Has well defined boundaries & assumptions
- Researchable
 - Can be accomplished using a scientific method
- Right balance between Research & Development
 - Manageable/feasible/practical
 - Fits the level of researcher's skill set
 - Time, resources, tools
- Low Risk Low Gain vs. High Risk High Gain
 - Is it free of unknown hazards/dangers?
- Ethical

Stating the problem

- Address following 3 questions in a suitable order
 - What are you planning to do?
 - Why is it important?
 - How are you planning to do it?
- Refine it as you understand the problem better

Research Objectives

- How do you plan to address your research problem?
- Summarize accomplishments you wish to achieve through the project
- Also, provides direction to the study
- Objective must be achievable
 - Set 3 to 5 objectives
 - Usually starts with To identify ..., To design ..., To develop ..., To analyse ..., To produce, ...
 - To produce a new dataset for automatic X-Ray image captioning
 - To design & develop a feature extraction method for automatic X-Ray image captioning
 - To design & develop a automatic image captioning system for lung X-Ray images
 - To enhance the performance of image captioning using object detection

Preliminary suppositions and implications

- Argue how and in what ways you believe your research will refine, revise, or extend existing knowledge in the subject area under investigation.
- Depending on the aims and objectives of your study, describe how the anticipated results will impact future scholarly research, theory, practice, forms of interventions, or policymaking.
- Note that such discussions may have either:
 - Substantive (a potential new policy) significance
 - Theoretical (a potential new understanding) significance, or
 - Methodological (a potential new way of analyzing) significance.

2. Literature Review

Literature review

Covered last week!

Research gap

- Connecting the related work to your research objectives

3. Proposed Method

Proposed Method

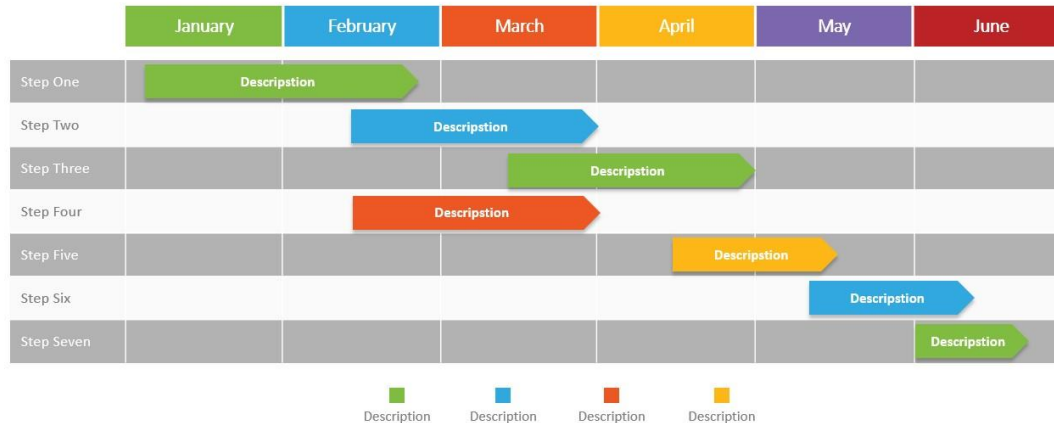
To be covered next week!

4. Schedule

Schedule/timeline

- A project timeline is a (visual) list of tasks or activities placed in chronological order
- View of the entirety of the project in one place.
- Provides an in-depth overview of the entire project from start to finish —and importantly, whether or not it's dependent on another task.

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5. Conclusion

Conclusion

- The conclusion reiterates the importance or significance of your proposal and provides a brief summary of the entire study.
 - Avoid repeating previous sentences
- This section should be only one or two paragraphs long, emphasizing why the research problem is worth investigating, why your research study is unique, and how it should advance existing knowledge.

6. Bibliography

Bibliography

Covered in a few more weeks!

References

- Slides modified from “Writing Research Proposals” by Hiroshi de Silva.
- Slides modified from “ITE1932 Technical Writing” - Nisansa De Silva
- “Fast phasing your thesis” - Dilum Bandara
- <https://gradcoach.com/what-is-a-research-proposal-dissertation-thesis/>